

Lecture 1

INTRODUCTION

- Syllabus is uploaded to my.nes.

- The textbook is:

Mas-Colell A., M. Whinston and J. Green, Microeconomic Theory, Oxford University Press, 1995.

We will cover Chapters 1,2,3 and 6.

- There will be weekly problem sets (20% of the grade), and a final (80%). Each problem set will be solved in the following week's TA seminar.
- This is a theory course; it will feel abstract occasionally, and that is natural. You are welcome to ask questions, request examples, graphs etc. for further clarification.
- In economics, we use mathematical models to analyze economic phenomena. Mathematics serves as a precise language to formulate issues and communicate ideas. Ultimately, the purpose is to make predictions related to economic variables, so that we can make sound decisions as policy makers or consultants.
- The first step is to understand the basic relations between economic variables. For example:
 - Suppose the price of commodity i increases. What happens to the market demand for good i , or that for some other good j ? What are the fundamental features of good i or j that may influence the outcome?
 - If a firm increases its own price, how will a rival firm respond?

These are basic theoretical questions that you need to understand as an economist, irrespective of whether you work as a consultant, a policy maker, or a researcher. A great part of your first-year study at NES will be about such basic questions, and the methods/tools that are used to study them.

- You will often hear that economists use unrealistic assumptions. But the real-life is extremely complicated. We need simplifying assumptions to make sense of it. Of course, we should avoid over-simplifications that can lead to misleading/unsatisfactory predictions, but the right criteria for that are context dependent. For example, if you are a policy maker who is trying to determine the optimal income tax rate for a given country, a highly sophisticated and realistic model could be useful. That is, you need a big model, with many parameters, and your parameters should be backed by data. But if you have a client who wants a short answer on a question between the market prices of the goods i and j , at the back of your mind what you need is a simple model that is able to convey the basic ideas forcefully. Stronger the assumptions easier will be the analysis...
- As a general rule, we should keep our models as simple as possible; realism by itself is not a good reason to relax an assumption. Rather, you should ask yourself what exactly you will gain in terms of predictive power from relaxing a particular assumption. If your predictions are invalidated empirically, then you should of course start checking which assumption may be responsible from that.
- Microeconomics focuses on small economic units such as individual consumers, firms, and their interactions, as opposed to the country-level analysis in macroeconomics.
- In Micro 1, our focus will be on individual consumers. We will study simple models of consumer behavior that provide a framework to address basic questions such as:
 - what sort of a consumption bundle may be chosen as a function of prices?
 - how would the choices change with the income of the consumer?
 - what sort of an insurance package may a consumer demand as a function of risk events, such as unemployment, health-risks etc.

Answering such questions for a particular market with great precision would require data-dependent methods. Our focus in this course will be on the gen-

eral theoretical framework in which these data-dependent/applied models are formulated. This general framework is referred to as *consumer theory*.

- In Micro 2, you will study firms' behavior. In Micro 3 and Game Theory, you will bring many consumers or firms together, and study their interactions in the marketplace. For now, we will focus on a single consumer, and the theoretical aspects of her choice behavior in the marketplace.

Preference and Choice (Chapter 1)

- X denotes the set of all alternatives in the environment. Depending on the choice problem that the decision maker faces, an “alternative” may refer to a basket of goods that can be purchased in a supermarket, a certain type of a job in the labor market, or an insurance policy...
- $B \subseteq X$ is a set of *feasible* alternatives; the so called *budget set*. For example, this may be all baskets of goods that our decision maker can purchase given his income; or all available jobs given his skill set.

How would this person (henceforth, the *agent*) make a choice from B ? In practice, there is a plethora of choice procedures that the agent can use, but we want a general approach that is easy to analyze and makes sense in many cases of interest.

Since we want a mathematical model, a natural approach is to think of the agent's choice problem as an optimization problem:

$$\max u(x) \quad \text{s.t.} \quad x \in B. \tag{P}$$

The objective function $u : X \rightarrow \mathbb{R}$ associates a number $u(x)$ with each alternative $x \in X$. The number $u(x)$ represents the satisfaction, or utility, that the agent experiences if she selects x . Hence, u is called a “utility function.”

Problem (P), then, tells us that the agent is trying to make a choice that maximizes her satisfaction. Intuitively, this makes perfect sense. By contrast, from an

empirical standpoint it does not mean much because we do not really know what “satisfaction/utility” means. Does it make sense to assume that it can be measured with a real valued function, u ? How can we test this assumption empirically?

All we observe is the agent’s choice behavior: She chooses x when y is available etc. Based on such observations, we (the analysts) can decide whether the agent prefers x over y , for any pair of alternatives x, y . Thereby, we are eliciting the agent’s preference relation, which is a relative concept of satisfaction that has empirical roots. Formally, a preference relation is a binary relation:

Definition 1. A *binary relation* R (on the set X) is a subset of the Cartesian product $X \times X$. We write $x R y$ to mean that $(x, y) \in R$, or in words, “ R relates x to y .”

A *preference relation* $\succsim$ is a binary relation that has a specific interpretation: $x \succsim y$ means “the agent thinks that x is as good as y ,” or equivalently, “the agent (weakly) prefers x to y .”

A utility function can now be defined as follows:

Definition 2. A function $u : X \rightarrow \mathbb{R}$ *represents* the preference relation $\succsim$ if for any $x, y \in X$ we have

$$x \succsim y \quad \Leftrightarrow \quad u(x) \geq u(y). \quad (1)$$

In this case, we say that u is a *utility function* that represents the agent’s preferences.

So, formally, a utility function is a numerical representation/formulation of agent’s relative preferences. Note that this an ordinal/order based concept: If u and v rank all the alternatives in the same way, then they represent the same preference relation. For example, the function $v = 2u$ represents the same preference relation as the function u , but they assign different numerical values/cardinals to a given alternative. It also follows that a utility function does not mean to measure satisfaction in an absolute/cardinal sense.

Going back to problem (P), remember that we are trying to find out the empirical content of the notion of a utility function. We now have an empirically observable preference relation, while a utility function refers to a numerical representation of it. Whether such a representation exists turns out to be equivalent to empirically testable properties of the preference relation.

Definition 3. A binary relation R is said to be

- *transitive* if for any $x, y, z \in X$

$$x R y \quad \& \quad y R z \quad \Rightarrow \quad x R z$$

- *complete* if for any $x, y \in X$, we have either $x R y$ or $y R x$.

For preference relations, transitivity is a consistency condition. It says that if x is preferred to y , and y to z , then x should be preferred to z . In turn, completeness means that the agent should never be indecisive; she should always be able to say whether x is better than y , or vice versa.

Note that the greater than or equal to relation, $\geq$, on the set of real numbers $\mathbb{R}$ is both transitive and complete. Hence, a preference relation that has a utility representation must satisfies these two properties. The next proposition shows that the converse is also true.

Proposition. *A binary relation $\succsim$ on a finite set X admits a utility representation $\Leftrightarrow$ it is complete and transitive.*

Proof. ($\Rightarrow$) This part easily follows from the fact that $\geq$ is a complete and transitive binary relation on $\mathbb{R}$.

($\Leftarrow$) Suppose $\succsim$ is complete and transitive. We need to construct a function $u : X \rightarrow \mathbb{R}$ that represents $\succsim$. For any $x \in X$, let $L(x) := \{z \in X : x \succsim z\}$, and define $u : X \rightarrow \mathbb{R}$ as $u(x) := |L(x)|$, where $|A|$ is the number of elements of a set $A \subseteq X$.

Pick a pair of points $x, y \in X$ with $x \succsim y$. For any $z \in X$ with $y \succsim z$ we also have $x \succsim z$ because $\succsim$ is transitive. Thus, $L(x) \supseteq L(y)$, and hence, $|L(x)| \geq |L(y)|$. This proves $\Rightarrow$ in statement (1).

To prove the converse implication in (1), suppose $u(x) \geq u(y)$, i.e., $|L(x)| \geq |L(y)|$ for some $x, y \in X$. Assume by contradiction that $x \succsim y$ is not true. Then $y \notin L(x)$. Moreover, since $\succsim$ is complete, “not $x \succsim y$ ” implies $y \succ x$. From the first part of the proof it then follows that $L(y) \supseteq L(x)$. We also have $y \in L(y)$ because $\succsim$ is complete. So, we have shown that $y \notin L(x) \subseteq L(y) \ni y$, implying that $L(x) \subsetneq L(y)$. Then $|L(x)| < |L(y)|$, which contradicts the assumption $|L(x)| \geq |L(y)|$. Hence, $|L(x)| \geq |L(y)|$ implies $x \succsim y$, as we seek. $\square$

A *strict preference relation* $\succ$ and an *indifference relation* $\sim$ can be derived from the weak preference $\succsim$ as follows:

$$\begin{aligned} x \succ y &\Leftrightarrow x \succsim y \ \& \ (\text{not } y \succsim x), \\ x \sim y &\Leftrightarrow x \succsim y \ \& \ y \succsim x. \end{aligned}$$

Problem set 1 contains some questions on the properties of these strict and weak preference relations.

DISCUSSION

Often economists say that a preference relation is “rational” if it is complete and transitive. Many experts on choice behavior, however, would say that indecisiveness/incompleteness of tastes is not really a form of bounded rationality. There is a growing interest in incomplete preferences in the behavioral economics literature, but in this course we abstract from this issue.

In fact, non-transitive preferences also arise naturally, especially in the context of social choice problems. Indeed, the following example shows that social preferences generated by the majority voting rule can be non-transitive.

Example (Condorcet Paradox). Consider a society with three individuals, $i = 1, 2, 3$. There are three possible social outcomes, a, b, c . (For example, each outcome may correspond to a candidate who is running for presidency.) So, the universal set of alternatives is $X = \{a, b, c\}$. The individuals’ preferences are as follows:

$$a \succ_1 b \succ_1 c, \quad c \succ_2 a \succ_2 b, \quad b \succ_3 c \succ_3 a.$$

The social preference relation $\succsim$ is defined as, for any $x, y \in X$,

$$x \succsim y \Leftrightarrow |\{i : x \succsim_i y\}| \geq |\{i : y \succsim_i x\}|.$$

That is, $x \succsim y$ iff the number of individuals who prefer x to y is greater than or equal to the number of individuals who prefer y to x . Note that individuals 1 and 2 (strictly) prefer a over b , and hence, $a \succ b$. Similarly, we have $b \succ c$ because individuals 1 and

3 prefer b over c . Finally, we also have $c \succ a$ because individuals 2 and 3 prefer c over a . To summarize, we get:

$$a \succ b \succ c \succ a.$$

That is, although $a \succ b$ and $b \succ c$, it is not true that $a \succ c$. Conclusion: The social preference $\succsim$ is not transitive.

Lecture 2

Continuity and Utility Representations in $\mathbb{R}^k$ (Chapter 3)

In the last lecture, we showed that the transitivity and completeness axioms are sufficient conditions for the existence of a utility function for a preference relation on a finite set X . In this lecture, we will focus on a possibly infinite set $X \subseteq \mathbb{R}^k$. We are interested in this environment essentially because we want to use differential calculus when solving optimization problems.

If X is countably infinite, a set of the form $\{x^1, x^2, \dots\}$, completeness and transitivity are still sufficient for the existence of a utility representation. (In this case, the logic behind the earlier existence proof remains valid. We just need to use a weighted counting method so that (i) $u(x) < \infty$ for every $x \in X$; and (ii) we can distinguish two infinite sets, $L(x)$ and $L(y)$ with $L(x) \subsetneq L(y)$.)

We can also cover a general set $X \subseteq \mathbb{R}^k$, but we need an additional, technical axiom. To motivate the axiom, consider a continuous utility function $u : X \rightarrow \mathbb{R}$. Take a sequence (x^n) in X that converges to a point $x \in X$. Then $u(x)$ is well defined, and we have $u(x) = \lim_n u(x^n)$ by continuity of u . Since the interval $[u(y), \infty)$ is a closed set, the following holds for any $y \in X$:

$$u(x^n) \geq u(y) \quad \forall n \quad \Rightarrow \quad u(x) \geq u(y).$$

Similarly, if $u(y) \geq u(x^n)$ for every n , then we have $u(y) \geq u(x)$. Thereby, we verified that the preference relation represented by u satisfies the following:

Definition 1. A preference relation $\succsim$ on a set $X \subseteq \mathbb{R}^k$ is said to be *continuous* if for any point $y \in X$ and any sequence (x^n) in X that converges

to a point $x \in X$ the following two properties hold:

$$\begin{aligned} x^n \succsim y \quad \forall n &\Rightarrow x \succsim y, \\ y \succsim x^n \quad \forall n &\Rightarrow y \succsim x. \end{aligned}$$

For any $y \in X$, the set $L(y) := \{z \in X : y \succsim z\}$ is called the *(weak) lower contour set* of y . Similarly, $U(y) := \{z \in X : z \succsim y\}$ is called the *(weak) upper contour set* of y . The strict lower and upper contour sets, respectively, are defined as $L_s(y) := \{z \in X : y \succ z\}$ and $U_s(y) := \{z \in X : z \succ y\}$.

Continuity means that weak preferences are preserved in the limit. Formally, it is equivalent to saying that for every $y \in X$, the sets $L(y)$ and $U(y)$ are relatively closed subsets of X . (“Relatively,” because we do not care about limit points that do not belong to X . For example, if $X = (0, 1]$, it is meaningless to talk about preferences at 0.)

Note that completeness of $\succsim$ implies $U_s(y) = X \setminus L(y)$ and $L_s(y) = X \setminus U(y)$. It is also known that a set is (relatively) open if and only if its complement is closed. It follows that $\succsim$ is continuous if and only if for every $y \in X$, the sets $U_s(y)$ and $L_s(y)$ are relatively open in X . This means that for every $y, z \in X$ with $z \succ y$ there exists an $\varepsilon > 0$ s.t.

$$\begin{aligned} z' \succ y \text{ for every } z' \in X \text{ with } \|z - z'\| < \varepsilon. \\ z \succ y' \text{ for every } y' \in X \text{ with } \|y - y'\| < \varepsilon. \end{aligned}$$

In words, $\succsim$ is continuous if and only if $z \succ y$ implies that z is strictly preferred to any alternative y' that is sufficiently close to y , and y is strictly worse than any alternative z' that is sufficiently close to z .

- The book uses a stronger, two-sided version of the continuity axiom (Definition 3.C.1). Your TAs will discuss that in their first seminar.

Unlike transitivity and completeness, the continuity axiom has no empirical content; it is not testable with (finite) data. Yet, it makes good sense, and it helps. The following is a classic theorem.

Theorem (Debreu). *A binary relation $\succsim$ on a set $X \subseteq \mathbb{R}^k$ is representable by a continuous utility function if and only if it is complete, transitive and continuous.*

We skip the proof of this theorem because it is too involved. Your TAs will give an example of a discontinuous preference relation that has no utility representation. For now, let us take a brief look into the preference relation in question, which is known as the *lexicographic preference*.

Example (Lexicographic preference). Denote by $x = (x_1, x_2)$, $y = (y_1, y_2)$ generic elements of $\mathbb{R}^2$. The lexicographic preference $\geq_L$ is defined as, for every $x, y \in \mathbb{R}^2$,

$$x \geq_L y \quad \Leftrightarrow \quad \text{either } x_1 > y_1, \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2).$$

It is easily checked that $\geq_L$ is complete and transitive. Figure 1 below shows that $\geq_L$ is not continuous. We can verify this using two equivalent definitions of continuity: (A) In the figure, the sequence (x^n) converges to x , we have $x^n >_L y$ for every n , but $y >_L x$. So, $U(y)$ is not closed. (B) The point x belongs to $L_s(y)$, but there is no ε -neighborhood around x that fits into $L_s(y)$ because any such set contains some x^n , which are all strictly better than y . Thus, $L_s(y)$ is not open.

Your TAs will show that $\geq_L$ cannot be represented by a utility function, and that is because of its discontinuity.

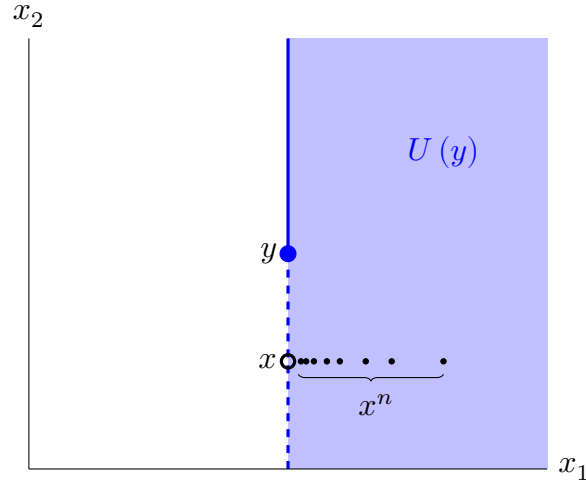


Figure 1: Discontinuity of lexicographic preference

Choice Rules and Rationalizability (Chapter 1 - C/D)

We have seen that the empirical content of representability by a utility function consists of transitivity and completeness axioms provided that we take a preference relation as the observable data. We have also noted that (relative) preferences of the agent can be elicited from choice data but we haven't clarified how. After all, what do completeness and transitivity mean in terms of the choice data? Let us now address these questions, for an arbitrary set of alternatives X .

Definition 2. A *choice structure* $(\mathcal{B}, C)$ on a set X consists of a collection $\mathcal{B}$ of nonempty subsets of X , and a set valued function C on $\mathcal{B}$ such that $C(B)$ is a nonempty subset of B for every $B \in \mathcal{B}$. We refer to C as a *choice correspondence* in general, and as a *choice function* provided that $C(B)$ contains a unique element for every $B \in \mathcal{B}$.

For any given set B , the subset $C(B)$ represents “chosen” or “choosable” alternatives from B . In general, multiple alternatives may be choosable. If

a unique alternative is chosen from each $B \in \mathcal{B}$, we have a choice function.

Given a binary relation $\succsim$ on X , let us use the following notation

$$\max(B, \succsim) := \{x \in B : x \succsim y \text{ for every } y \in B\}.$$

This is the set of best elements in B according to $\succsim$. (The textbook writes $C^*(B, \succsim)$ in place of $\max(B, \succsim)$.)

Consider an agent, and suppose his observed choices are depicted by a choice correspondence C . When can we say that this agent behaves as if he has a utility function? We have seen that this is equivalent to seeking a complete and transitive preference relation. The question, then, becomes whether we can find a complete and transitive binary relation $\succsim$ on X s.t.

$$C(B) = \max(B, \succsim) \quad \text{for every } B \in \mathcal{B}.$$

If such a binary relation $\succsim$ exists, we say that C is *rationalizable*, or $\succsim$ *rationalizes* C . So, we want to understand what sort of choice correspondences are rationalizable. Are there testable properties of a choice correspondence that characterize its rationalizability?

Consider the following axiom toward an answer:

Weak Axiom of Revealed Preferences (WARP). If $y \in B$ and $x \in C(B)$ for some $B \in \mathcal{B}$, then for any $B' \in \mathcal{B}$ with $x \in B'$ and $y \in C(B')$, we must also have $x \in C(B')$.

The following definition will help interpret this axiom:

Definition 3. Let us say that x is *revealed preferred* to y , denoted $x \succsim^* y$, if there exists a set $B \in \mathcal{B}$ s.t. $y \in B$ and $x \in C(B)$.

In words, x is revealed preferred to y means that x is chosen from at least one budget set where y is also available.

Similarly, if $x \in C(B)$ and $y \in B \setminus C(B)$, we can infer that “ x is strictly revealed preferred to y .”

Given these definitions, WARP amounts to saying that if x is revealed preferred to y , then y cannot be strictly revealed preferred to x . That is, if $x \succsim^* y$, and y is chosen from some set that contains x , then x must also be chosen from the latter set. This consistency condition turns out to be equivalent to rationalizability of a choice correspondence. We will prove that characterization in the next lecture.

Lecture 1

Review

Shortly put, modern economics consists of optimization problems. The following is a list of common examples:

- A consumer maximizes her utility function subject to a budget constraint.
- A producer maximizes its profit subject to technological constraints.
- A social planner maximizes the welfare of the society subject to resource constraints.

The first part of this course is devoted to static optimization problems. We will typically assume that the objective function and the constraint functions are differentiable. Our focus will be on the role of first order conditions (FOCs), how and when we can use them. To that end, in this lecture we will review the notion of a derivative, and some related concepts.

Some Basic Notation and Definitions

- $\forall$ means “for every.”
- $\exists$ means “there exists” or “for some.”
- iff means “if and only if”
- $\mathbb{R}^n$ denotes the n -dimensional Euclidean space with generic elements $x = (x_i)$, $y = (y_i)$ etc.
- $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$ gives the distance between x and the origin, 0. (This is the norm of x .)

- $\|x - y\| = \left(\sum_{i=1}^n (x_i - y_i)^2\right)^{1/2}$ gives the distance between x and y .
- A sequence (x^k) converges to x , written $x^k \rightarrow x$ or $\lim x^k = x$, if $\forall \varepsilon > 0, \exists$ an index K such that $k > K$ implies $\|x^k - x\| < \varepsilon$. That is, the distance between x^k and x becomes arbitrarily small for sufficiently large indices. (See Figure 1.1.)

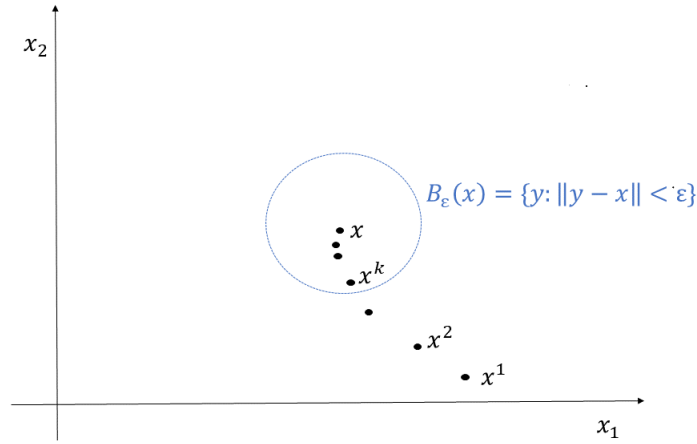


Figure 1.1: Convergence

- When $n = 1$, we have $\|x\| = (x^2)^{1/2} = |x|$. So, a sequence of real numbers x^k converges to a number x iff $|x^k - x| \rightarrow 0$.
- A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous at a point $x \in \mathbb{R}^n$ if for any sequence x^k in $\mathbb{R}^n$ that converges to x , we have $\lim f(x^k) = f(x)$. Equivalently, f is continuous at x if $\forall \varepsilon > 0, \exists \delta > 0$ such that $\|y - x\| < \delta$ implies $\|f(y) - f(x)\| < \varepsilon$. That is, $f(y)$ becomes arbitrarily close to $f(x)$ whenever y is sufficiently close to x . Put differently, when we replace x with nearby points, the f value does not change radically. We say that f is continuous if it is continuous at any point

on its domain. This implies that the graph of f that depicts the pairs $(x, f(x))$ is a connected object with no holes/jumps. A continuous, real-valued function on a compact subset of $\mathbb{R}^n$ is guaranteed to have a maximum and a minimum. (Weierstrass Theorem.)

- We write $\lim_{y \rightarrow x} f(y) = A$ to mean that $f(y^k) \rightarrow A$ for any sequence $y^k \rightarrow x$ (with $y^k \neq x$).
- For a function of one variable, we write $\lim_{y \rightarrow x-} f(y) = A$ to mean that $f(y^k) \rightarrow A$ for any sequence $y^k \rightarrow x$ with $y^k < x$. Similarly, $\lim_{y \rightarrow x+} f(y) = A$ means $f(y^k) \rightarrow A$ for any sequence $y^k \rightarrow x$ with $y^k > x$. These are the left and right limits, respectively.
- The definition of convergence (or divergence) to ∞ is suitably modified: $\lim_{y \rightarrow x} f(y) = \infty$ means that $f(y^k)$ becomes arbitrarily large as $y^k \rightarrow x$.

Derivatives and Differentiability

Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$.

- The derivative of f at a point x , denoted $f'(x)$, is defined as follows:

$$f'(x) = \lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x}. \quad (1)$$

Intuitively, this is the slope of f at x . More formally, $f'(x)$ is the slope of the line that is tangent to the graph of f at the point $(x, f(x))$. (See Figure 1.2.)

- In general, the limit on the right hand side of (1) may not be well-defined because for distinct sequences $y^k \rightarrow x$, the ratio $\frac{f(y^k) - f(x)}{y^k - x}$ may converge to

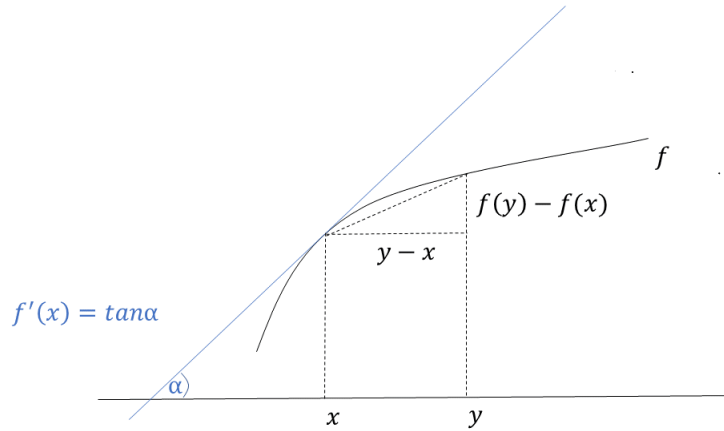


Figure 1.2: Derivative

distinct numbers. In this case, f is not differentiable at x . Otherwise, $f'(x)$ is well-defined, and we say that f is differentiable at x . In turn, if f is differentiable at every point on its domain, we simply say that f is differentiable.

- One can also be interested in left and right derivatives:

$$f'(x-) = \lim_{y \rightarrow x-} \frac{f(y) - f(x)}{y - x} \quad \text{and} \quad f'(x+) = \lim_{y \rightarrow x+} \frac{f(y) - f(x)}{y - x}.$$

By definition, if $f'(x)$ exists, then the left and right derivatives are equal to each other. But if the function has a kink at a point x , then the left and right derivatives can be distinct. (See Figure 1.3.)

- Typically, we will assume full differentiability, and that the derivative is a finite number. But for points that lie on the boundary of a set (typically, the origin), we may seek only a single directional derivative—we cannot approach the point from the other direction anyway—and allow that directional derivative to be infinite.

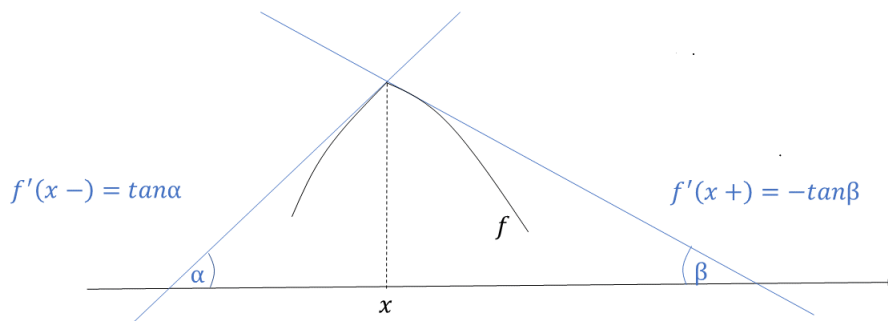


Figure 1.3: Left and Right Derivatives

The Role of Differentiation in Optimization

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at a point x , and that $f'(x) = a$ is a (strictly) positive number. Then for any y' that is close to but greater than x ,

$$a = \lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x} \Rightarrow f(y') - f(x) \simeq a(y' - x) > 0.$$

Hence, $f(y') > f(x)$ for such y' , and this means x does not maximize f . Similarly, if $f'(x) = a < 0$, then $f(y') > f(x)$ for y' that is close to but smaller than x .

Conclusion. If x is a local maximizer of a differentiable function f (meaning there exists an $\varepsilon > 0$ such that $f(x) \geq f(y)$ for any $y \in B_\varepsilon(x) = \{y : |y - x| < \varepsilon\}$), then $f'(x) = 0$. (A necessary first order condition.) Similarly, if x is a local minimizer, then we also have $f'(x) = 0$.

- First order conditions do not distinguish between minimizers and maximizers, or saddle points for that matter. (For example, 0 is a saddle point of $f = x^3$.)

As such, first order conditions are necessary, but not sufficient for maximization or minimization.

- Second order conditions can be sufficient to identify *local* optima. For example, if $f'(x^*) = 0$ and $f''(x^*) < 0$ for a twice differentiable function f , then x^* is guaranteed to be a (strict) local maximum. (See Figure 1.4.)

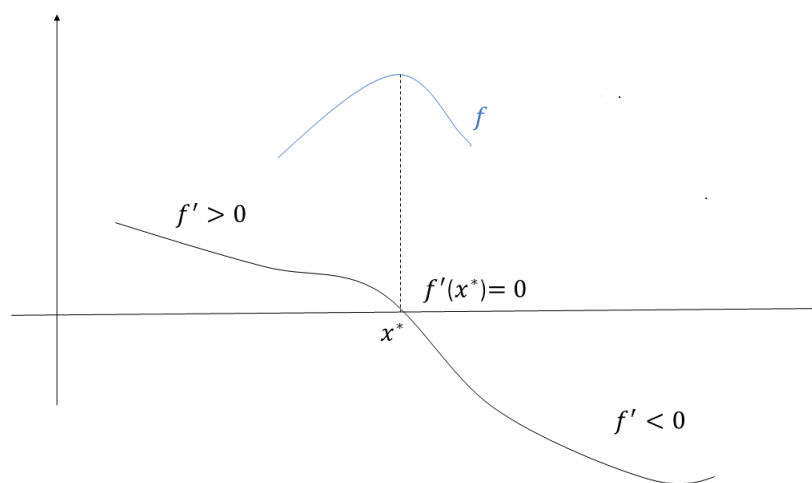


Figure 1.4: Second Order Conditions for Unconstrained Maximization

Multivariate Functions

Let $S \subseteq \mathbb{R}^n$ be an open set (meaning $\forall x \in S, \exists \varepsilon > 0$ such that $B_\varepsilon(x) = \{y : \|y - x\| < \varepsilon\} \subseteq S$), and consider a function $f : S \rightarrow \mathbb{R}$.

Definition. Given a point $x \in S$, and an arbitrary vector $h \in \mathbb{R}^n$, the derivative of f at x in the direction h , denoted $Df(x; h)$, is given by

$$Df(x; h) = \lim_{t \rightarrow 0+} \frac{f(x + th) - f(x)}{t}.$$

This measures the rate of change of f between the points x and $x + h$ near x , but it also depends on the length of the vector h , among other things. (More on this in equation (2) below.)

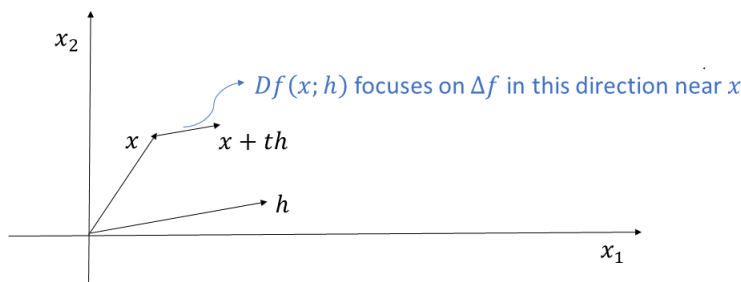


Figure 1.5: On the Directional Derivative

- As before, $Df(x; h) > 0$ implies $f(x + th) > f(x)$ for small $t > 0$, which rules out the possibility of x being a local maximizer.

Definition. Let $e_j = (0, \dots, 0, 1, 0, \dots, 0)$ be the j th coordinate vector. If it exists, the j th partial derivative of f at the point x , denoted $\frac{\partial f(x)}{\partial x_j}$ is defined as:

$$\frac{\partial f(x)}{\partial x_j} = \lim_{t \rightarrow 0} \frac{f(x + te_j) - f(x)}{t}.$$

This is equivalent to saying

$$\frac{\partial f(x)}{\partial x_j} = Df(x; e_j) = -Df(x; -e_j).$$

Conclusion. Suppose $x \in S$ is a local maximizer of f (meaning there exists an $\varepsilon > 0$

such that $f(x) \geq f(y)$ for any $y \in B_\varepsilon(x)$). Assume further that all partial derivatives of f exist at the point x . Then we must have $\frac{\partial f(x)}{\partial x_j} = 0$ for each j . (Necessary first order conditions in unconstrained optimization. Same conditions must hold at a local minimum.)

A General Definition

Towards a general (i.e., direction-free) definition of a derivative, let us first go back to the univariate case. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at x , and set $a = f'(x)$. Define a linear function $g(y) = ay + b$, where b is such that $g(x) = f(x)$, meaning $b = f(x) - ax$. (See Figure 1.6.)

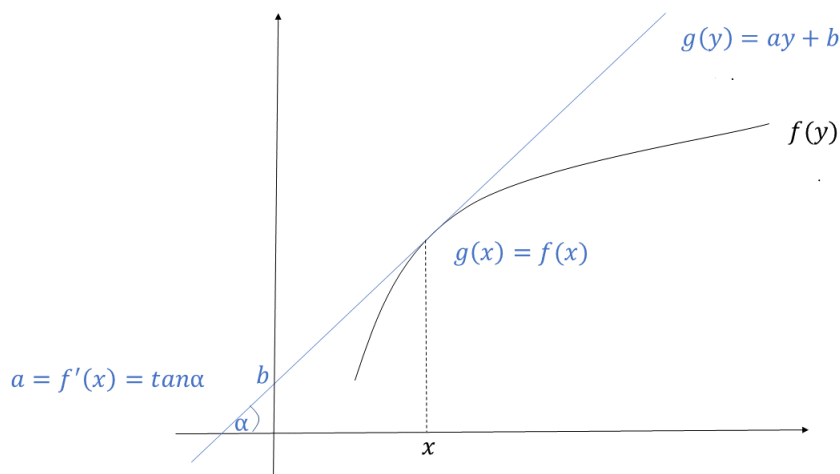


Figure 1.6: Derivative as a Linear Approximation

By definitions, the graph of the linear function g is tangent to that of f at the point x . So, we have $g(y) \simeq f(y)$ near x . Since we also have $g(x) = f(x)$, it also follows that $g(y) - g(x) \simeq f(y) - f(x)$ near x . Thus, $g(y) - g(x) = a(y - x)$ approximates the actual change $f(y) - f(x)$.

Indeed,

$$\begin{aligned}
f'(x) &= \lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x} \Rightarrow \\
0 &= \lim_{y \rightarrow x} \frac{y - x}{|y - x|} \left(\frac{f(y) - f(x)}{y - x} - f'(x) \right) \\
&= \lim_{y \rightarrow x} \frac{f(y) - f(x)}{|y - x|} - f'(x) \frac{y - x}{|y - x|} \\
&= \lim_{y \rightarrow x} \frac{f(y) - f(x) - f'(x)(y - x)}{|y - x|}.
\end{aligned}$$

Thus, $f'(x)(y - x)$ approximates $f(y) - f(x)$, and the approximation is so good that the difference between the two expressions is approximately 0 even after we divide it by the small number $|y - x|$.

A multivariate derivative is defined analogously:

Definition. Let $S \subseteq \mathbb{R}^n$ be an open set. If it exists, the derivative of a function $f : S \rightarrow \mathbb{R}^m$ at a point $x \in S$, denoted $Df(x)$, is the $m \times n$ matrix A such that

$$\lim_{y \rightarrow x} \frac{f(y) - f(x) - A(y - x)}{\|y - x\|} = 0.$$

In what follows, f denotes a multivariate function as in the definition above. Since f has m -dimensional values, we can think of this function as a collection of m functions, $f_1, \dots, f_m$, each mapping S into $\mathbb{R}$. That is, $f(y) = (f_1(y), \dots, f_m(y))$ for each $y \in S$.

- By definition, $Df_j(x)$ is a $1 \times n$ vector for each j . It can actually be shown that

$$Df(x) = \begin{bmatrix} Df_1(x) \\ \cdot \\ \cdot \\ \cdot \\ Df_m(x) \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \cdot & \cdot & \cdot & \frac{\partial f_1(x)}{\partial x_n} \\ \cdot & \cdot & & & \cdot \\ \cdot & & \cdot & & \cdot \\ \cdot & & & \cdot & \cdot \\ \frac{\partial f_m(x)}{\partial x_1} & \cdot & \cdot & \cdot & \frac{\partial f_m(x)}{\partial x_n} \end{bmatrix}$$

- In particular, when $m = 1$ so that f itself is real-valued, we have

$$Df(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} & \cdot & \cdot & \cdot & \frac{\partial f(x)}{\partial x_n} \end{bmatrix}.$$

Moreover, for any $h \in \mathbb{R}^n$, we have

$$Df(x; h) = Df(x)h. \tag{2}$$

Exercise. Prove the equation (2) by letting $y = x + th$ in the definition of $Df(x)$.

Lectures 2-3

Optimization Problems with Equality Constraints

Chapter 5

Given an open set $S \subseteq \mathbb{R}^n$, let f and $g_1, \dots, g_k$ be functions that map S into $\mathbb{R}$. Consider the following problem:

$$\max_{x \in S} f(x) \quad \text{s.t.} \quad g_i(x) = 0 \text{ for } i = 1, \dots, k. \quad (\text{P})$$

If we set $g(x) = (g_1(x), \dots, g_k(x))$, and $\mathcal{D} = \{x \in S : g(x) = 0\}$, this problem can simply be stated as:

$$\max f(x) \quad \text{s.t.} \quad x \in \mathcal{D}. \quad (\text{P})$$

$x^* \in \mathcal{D}$ is a local solution to this problem if there exists an $\varepsilon > 0$ s.t. $f(x^*) \geq f(x)$ for any $x \in B_\varepsilon(x^*) \cap \mathcal{D}$.

To be able to use the FOCs, the constraints must satisfy a technical condition:

Definition. The constraint qualification (CQ) holds at the point x^* if the $k \times n$ matrix $Dg(x^*)$ has rank k . This is equivalent to saying that the vectors $Dg_1(x^*), \dots, Dg_k(x^*)$ are linearly independent, meaning $\sum_{i=1}^k \lambda_i Dg_i(x^*) = 0$ implies $\lambda_i = 0$ for each i .

- Recall that an n -dimensional space can have at most n linearly independent vectors. Thus the CQ can hold only if $n \geq k$. That is, the number of constraints should not exceed the number of variables.

The following is our first major theorem.

Theorem (Lagrange Theorem). *Suppose that in problem (P) the functions f and $g_1, \dots, g_k$ are continuously differentiable (C^1). Assume further that $x^* \in \mathcal{D}$ is a local*

solution to this problem that satisfies the constraint qualification. Then there exists a vector $\lambda^* = (\lambda_1^*, \dots, \lambda_k^*)$ such that

$$Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg_i(x^*) = 0. \quad (1)$$

That is, $\frac{\partial f(x^*)}{\partial x_j} + \sum_{i=1}^k \lambda_i^* \frac{\partial g_i(x^*)}{\partial x_j} = 0$ for $j = 1, \dots, n$.

- Let $L(\lambda, x) = f(x) + \sum_{i=1}^k \lambda_i g_i(x)$. This is the so called “Lagrangian function.” Then condition (1) amounts to saying $\frac{\partial L(\lambda^*, x^*)}{\partial x_j} = 0$ for each j , while the constraint $g_i(x^*) = 0$ is equivalent to the condition $\frac{\partial L(\lambda^*, x^*)}{\partial \lambda_i} = 0$. So, at a local solution that satisfies the CQ, all partial derivatives of L must be zero.
- A point x^* minimizes f on a subset of S iff it maximizes $-f$ on that set. Thus, FOCs for minimization problems can be derived from (1) by replacing f with $-f$. Since the theorem does not comment on the sign of the multipliers, $\lambda_1^*, \dots, \lambda_k^*$, there is actually no need to change the theorem statement for minimization problems.
- FOCs (1) give us n equations in $n + k$ unknowns, $(\lambda_1^*, \dots, \lambda_k^*, x_1^*, \dots, x_n^*)$. We also have k constraints. Altogether, there are $n + k$ equations and $n + k$ unknowns.

Geometric Meaning

Suppose we have a single constraint g . Then the Lagrange Theorem simply says:

$$Df(x^*) = -\lambda^* Dg(x^*) \quad \exists \lambda^* \in \mathbb{R}. \quad (2)$$

That is, the vectors $Df(x^*)$ and $Dg(x^*)$ are parallel to each other.

The level curve of f that goes through the point x^* collects all $x \in S$ such that $f(x) = f(x^*)$; this is the set $\{x \in S : f(x) = f(x^*)\}$. It can be shown that

$Df(x^*)$ is perpendicular to the hyperplane that is tangent to this level curve. That is, the tangent hyperplane equals $T = \{x \in \mathbb{R}^n : x \cdot Df(x^*) = 0\}$, where $\cdot$ denotes the dot product ($x \cdot y = \sum_{j=1}^n x_j y_j$). Put differently, $Df(x^*)$ is the “normal” of the tangent hyperplane. Similarly, $Dg(x^*)$ is the normal of the hyperplane that is tangent to the level curve of g that goes through x^* . Since $g(x^*) = 0$, this level curve equals $\{x \in S : g(x) = 0\}$, and the corresponding tangent hyperplane equals $T' = \{x \in \mathbb{R}^n : x \cdot Dg(x^*) = 0\}$.

Note that the parallelity condition (2) implies $T = T'$. This means that the two level curves are actually tangent to each other. (See Figure 2.1.)

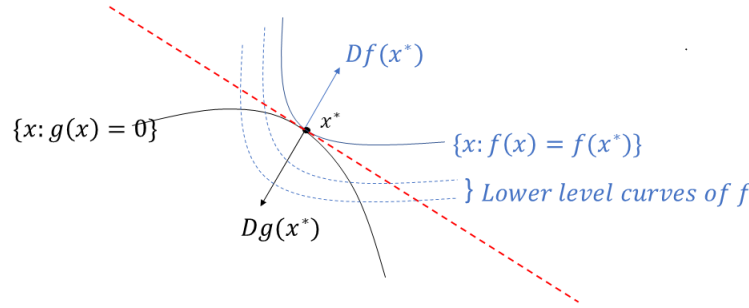


Figure 2.1: First Order Conditions with a Single Equality Constraint

Example 1 (Consumer’s Problem). Let $p = (p_j) \in \mathbb{R}^n$ be a strictly positive price vector, where p_j represents the price of the j th commodity. A commodity bundle is a strictly positive vector $x \in \mathbb{R}_{++}^n$. The cost of such a bundle is $px = p \cdot x = \sum_{j=1}^n p_j x_j$. Let $w > 0$ be the consumer’s wealth. The consumer cannot select a bundle that costs more than her wealth, so she faces a budget constraint $g(x) = w - px = 0$. Let $f : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$ be a utility function that represents the consumer’s tastes over

consumption bundles. Then the consumer's problem is

$$\max f(x) \quad \text{s.t.} \quad g(x) = 0 \text{ and } x_j > 0 \text{ for each } j.$$

Since strict inequality constraints do not influence FOCs, effectively we have only one constraint here, $g(x) = 0$. Note that $Dg(x) = -p$. Since $p \neq 0$ and a nonzero vector is linearly independent by itself, it follows that the CQ holds at any relevant point. The FOCs then say

$$Df(x^*) = -\lambda^* Dg(x^*) = \lambda^* p \quad \exists \lambda^* \in \mathbb{R}.$$

(See Figure 2.2.)

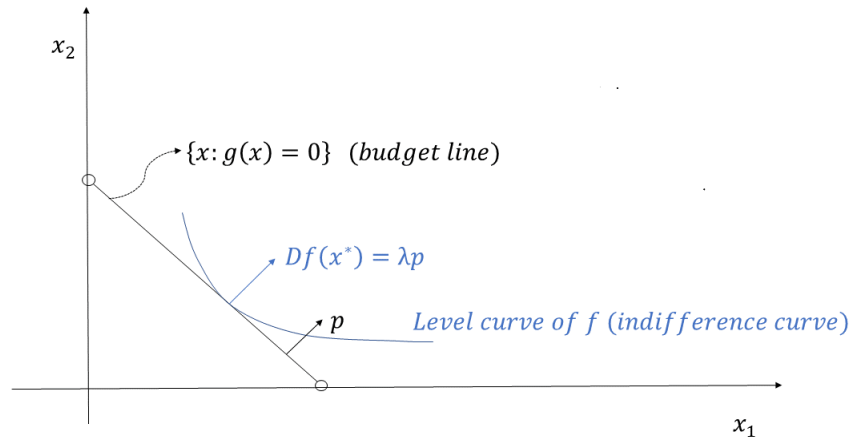


Figure 2.2: First Order Conditions in Consumer's Problem

Remark. Problem Set 1 contains a more general form of the consumer's problem with weak inequality constraints.

The Role of the Constraint Qualification

The CQ often holds in economic problems, but in this course we will not take this for granted. If the CQ fails, the derivatives of the constraints are linearly dependent. Consequently, all linear combinations of these vectors may lie in a small subspace. In this case, equation (1) may have no solution in $\lambda_1, \dots, \lambda_k$. That is, the FOCs may fail at a local solution. (See Figure 2.3.)

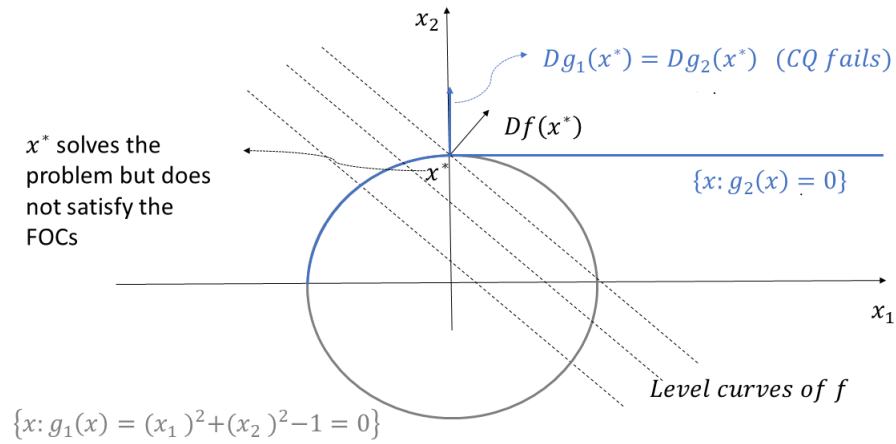


Figure 2.3: Failure of the Constraint Qualification and FOCs

Here is a further example with a single constraint:

Example 2. Consider the following problem

$$\max_{(x,y) \in \mathbb{R}^2} -y \quad \text{s.t.} \quad g(x,y) = y^3 - x^2 = 0.$$

Note that the constraint implies $y \geq 0$. Thus, the unique solution to the problem is given by $(x^*, y^*) = (0, 0)$. However, $Dg(x^*, y^*) = (0, 0)$, so the CQ fails at the solution. Meanwhile, $Df(x^*, y^*) = (0, -1)$, which cannot be expressed as $\lambda Dg(x^*, y^*)$. Thus, the FOCs fail at the solution.

- As before, FOCs are not sufficient to identify a local solution. (See Figure 2.4).

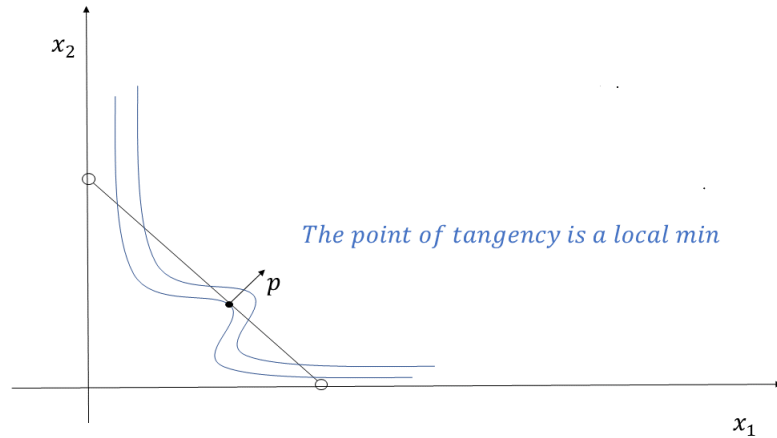


Figure 2.4: Consumer's Problem with a Local Min

The Cookbook Procedure

We have seen that FOCs can at best help identify local solutions, and they are not sufficient even to that end. Still, under certain conditions, we can use FOCs systematically to identify the global solutions of an optimization problem. This is how:

Step 1. Check if the optimization problem has a (global) solution. (Weierstrass Thm.) If yes, go to:

Step 2. Check if the CQ holds at all potential solutions x^* . If yes, go to:

Step 3. Find all critical points: Those points that satisfy all constraints as well as the FOCs (1) for some $\lambda_1^*, \dots, \lambda_k^*$.

Step 4. Select the best critical points: If $x^1, \dots, x^m$ are the critical points from Step 3, in a maximization problem select $x^{l'}$ s.t. $f(x^{l'}) \geq f(x^l)$ for $l = 1, \dots, m$.

- We have seen that if Step 2 fails, then the problem may have a solution that does not appear among critical points. In turn, if Step 1 fails, then you may mistakenly select a critical point that does not actually solve the problem. (See Figure 2.5.)

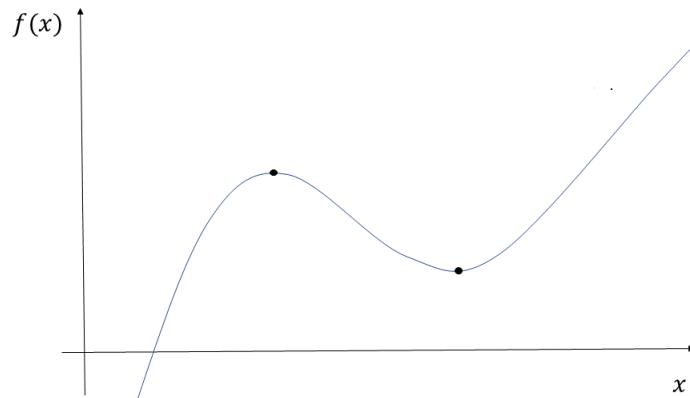


Figure 2.5: Two Critical Points with no Global Optima

- If the CQ fails at a few special points, you can simply add them to your set of critical points.

Lagrange Multipliers as Shadow Prices

In problem (P), a Lagrange multiplier λ_i^* measures how much the maximum value of the objective function f would increase if we were to relax the i th constraint

slightly. In this sense, the i th multiplier is the marginal cost of a constraint (called a “shadow price”).

To prove this, let us fix a constraint i_o , and modify the problem (P) as follows:

$$\max_{x \in S} f(x) \quad \text{s.t.} \quad g_{i_o}(x) + c = 0, \text{ and } g_i(x) = 0 \text{ for } i \neq i_o.$$

Here, $c \in \mathbb{R}$ is a parameter. We would like to understand how the maximum value changes as a function of c .

Let $x^*(c)$ be a solution, and suppose this defines a differentiable function in c . Define

$$V(c) = f(x^*(c)).$$

By the Chain rule,

$$\frac{dV(c)}{dc} = \sum_{j=1}^n \frac{\partial f(x^*(c))}{\partial x_j} \frac{\partial x_j^*(c)}{\partial c}. \quad (3)$$

Let $\lambda_i^*(c)$ be the i th multiplier, as a function of c . By FOCs (1), equation (3) can be written as

$$\begin{aligned} \frac{dV(c)}{dc} &= \sum_{j=1}^n \left(- \sum_{i=1}^k \lambda_i^*(c) \frac{\partial g_i(x^*(c))}{\partial x_j} \right) \frac{\partial x_j^*(c)}{\partial c} \\ &= \sum_{i=1}^k - \lambda_i^*(c) \sum_{j=1}^n \frac{\partial g_i(x^*(c))}{\partial x_j} \frac{\partial x_j^*(c)}{\partial c} \end{aligned} \quad (4)$$

Moreover, for all $c \in \mathbb{R}$ we have

$$g_{i_o}(x^*(c)) + c = 0, \text{ and } g_i(x^*(c)) = 0 \text{ for } i \neq i_o.$$

Differentiating these functional equalities w.r.t. c yield

$$\sum_{j=1}^n \frac{\partial g_{i_o}(x^*(c))}{\partial x_j} \frac{\partial x_j^*(c)}{\partial c} = -1, \text{ and } \sum_{j=1}^n \frac{\partial g_i(x^*(c))}{\partial x_j} \frac{\partial x_j^*(c)}{\partial c} = 0 \text{ for } i \neq i_o.$$

Finally, if we substitute these equalities into (4) we get the following:

$$\begin{aligned}\frac{dV(c)}{dc} &= -\lambda_{i_o}^*(c) \times (-1) + \sum_{i \neq i_o}^k -\lambda_i^*(c) \times 0 \\ &= \lambda_{i_o}^*(c)\end{aligned}$$

Conclusion. $\frac{dV(c)}{dc} = \lambda_{i_o}^*(c)$. That is, the rate of change of the maximum value w.r.t. c equals the multiplier of the corresponding constraint, i_o .

Proof of the Lagrange Theorem

Let $x^* \in \mathcal{D}$ be a local solution to the problem (P) that satisfies the CQ. If $n = k$, there is nothing to prove, for then we have n linearly independent vectors in $\mathbb{R}^n$, $Dg_1(x^*), \dots, Dg_n(x^*)$, so any other vector can be expressed as a linear combination of these. Therefore, without loss of generality we can assume $n > k$.

By the CQ, the rows of the $k \times n$ matrix $Dg(x^*)$ are linearly independent. Therefore, this matrix also has k linearly independent columns.¹ By reordering the coordinates $x_1, \dots, x_n$ if necessary, let us assume that the first k columns of the matrix $Dg(x^*)$ are linearly independent.

In place of x , let us write (y, z) where y collects the first k coordinates of x , and z collects the remaining $n - k$ coordinates. Accordingly, we denote by $D_y g(y, z)$ the $k \times k$ matrix that consists of the derivatives of the constraints w.r.t. to the first k

¹This is a fundamental theorem in linear algebra, which can equivalently be stated as “the row rank of a matrix equals its column rank.” Thanks to this theorem, we can talk about “the rank” of a matrix, which refers to the maximum number of linearly independent columns and rows. If you are interested, this page offers a quick read on the rank of a matrix: [https://en.wikipedia.org/wiki/Rank_\(linear_algebra\)#Proofs_that_column_rank=row_rank](https://en.wikipedia.org/wiki/Rank_(linear_algebra)#Proofs_that_column_rank=row_rank)

variables. That is,

$$D_y g(y, z) = \begin{bmatrix} \frac{\partial g_1(y, z)}{\partial y_1} & \cdot & \cdot & \cdot & \frac{\partial g_1(y, z)}{\partial y_k} \\ \cdot & \cdot & & & \cdot \\ \cdot & & \cdot & & \cdot \\ \cdot & & & \cdot & \cdot \\ \frac{\partial g_k(y, z)}{\partial y_1} & \cdot & \cdot & \cdot & \frac{\partial g_k(y, z)}{\partial y_k} \end{bmatrix}.$$

$D_z g(y, z)$, $D_y f(y, z)$ etc. are defined analogously.

Since $x^* = (y^*, z^*)$ satisfies the constraints, we have $g(y^*, z^*) = 0$. Moreover, $D_y g(y^*, z^*)$ is an invertible matrix by construction. Thus, by the Implicit Function Theorem, we can use the equation $g(y, z) = 0$ to solve y as a function of z in a neighborhood of z^* . That is, there exist an open set Z in $\mathbb{R}^{n-k}$ that contains z^* and a C^1 function $h : Z \rightarrow \mathbb{R}^k$ such that $h(z^*) = y^*$, while $(h(z), z) \in S$ and $g(h(z), z) = 0$ for each $z \in Z$.

Set $f^*(z) = f(h(z), z)$ for $z \in Z$, and note that z^* is a local maximum of the function f^* on the set Z . Indeed, by assumption, there exists an $\varepsilon > 0$ s.t. $f^*(z^*) = f(x^*) \geq f(x) = f(y, z)$ for any $(y, z) \in B_\varepsilon(x^*) \cap \mathcal{D}$. The points $(y, z) = (h(z), z)$ satisfy the constraint $g(y, z) = 0$ by definition of h . That is, $(h(z), z) \in \mathcal{D}$ for every $z \in Z$. Moreover, since h is continuous and $h(z^*) = y^*$, there exists a neighborhood Z' of z^* such that $(h(z), z) \in B_\varepsilon(y^*, z^*) = B_\varepsilon(x^*)$ for every $z \in Z'$. Thus, it follows that $f^*(z^*) \geq f(h(z), z) = f^*(z)$ for all $z \in Z'$.

As z^* is an unconstrained local maximizer of the function f^* , the FOCs yield $Df^*(z^*) = 0$. Hence, by the chain rule, we have

$$0 = \frac{\partial f^*(z^*)}{\partial z_j} = \sum_{l=1}^k \frac{\partial f(h(z^*), z^*)}{\partial y_l} \frac{\partial h_l(z^*)}{\partial z_j} + \frac{\partial f(h(z^*), z^*)}{\partial z_j} \quad \text{for } j = 1, \dots, n-k.$$

In matrix form, this simply means

$$0 = Df^*(z^*) = D_y f(h(z^*), z^*) Dh(z^*) + D_z f(h(z^*), z^*).$$

Equivalently,

$$D_z f(h(z^*), z^*) = -D_y f(h(z^*), z^*) Dh(z^*) \quad (5)$$

By the implicit function theorem, we have

$$Dh(z^*) = -D_y g(y^*, z^*)^{-1} D_z g(y^*, z^*). \quad (6)$$

Substitute (6) into (5), to get

$$D_z f(h(z^*), z^*) = D_y f(h(z^*), z^*) D_y g(y^*, z^*)^{-1} D_z g(y^*, z^*).$$

Thus, if we let $\lambda^* = -D_y f(h(z^*), z^*) D_y g(y^*, z^*)^{-1}$, we get

$$D_z f(h(z^*), z^*) = -\lambda^* D_z g(y^*, z^*).$$

So, the FOCs with respect to the last $n - k$ variables hold with this definition of λ^* .

As for the remaining k variables, simply note that by definition of λ^* , we have

$$-\lambda^* D_y g(y^*, z^*) = D_y f(h(z^*), z^*),$$

as we seek. □

- To prove the equality (6), differentiate the equality $g(h(z), z) = 0$ with respect

to z and evaluate at z^* . This yields

$$D_y g(h(z^*), z^*) Dh(z^*) + D_z g(h(z^*), z^*) = 0.$$

Then we can premultiply this equation with $D_y g(h(z^*), z^*)^{-1}$.

More on the Consumer's Problem

In Example 1 above, the constraint set is given by

$$\mathcal{D}_0 = \{x \in \mathbb{R}^n : w - px = 0 \text{ and } x_j > 0 \text{ for } j = 1, \dots, n\}.$$

This set is not closed, and consequently we cannot use the Weierstrass theorem to guarantee the existence of a solution. Intuitively, the problem here is that the consumer may not want to spend money to some commodities, which can lead to corner “solutions” with some zero coordinates. While such points are left out of the domain of the choice problem in Example 1, we did so simply to avoid weak inequalities in line with the Lagrange Theorem.

Instead, the classical (and more general) form of the consumer's problem is defined by weak inequalities:

$$\max f(x) \quad \text{s.t.} \quad w - px \geq 0 \text{ and } x_j \geq 0 \text{ for } j = 1, \dots, n.$$

Let $g_j(x) = x_j$ for $j = 1, \dots, n$ and $g_{n+1}(x) = w - px$. Accordingly, the constraint set is defined by $n + 1$ weak inequalities:

$$\mathcal{D} = \{x \in \mathbb{R}^n : g_i(x) \geq 0 \text{ for } i = 1, \dots, n + 1\}.$$

So, the problem now is:

$$\max f(x) \quad \text{s.t.} \quad x \in \mathcal{D}.$$

If p is strictly positive, then the set $\mathcal{D}$ is compact, and a solution exists by the Weierstrass theorem.

As noted earlier, in general, a solution x^* may as well have some zero coordinates, and in that cases we have a genuine inequality constraint problem that requires a different set of tools. We will address such problems in the next lecture.

However, if $x_j = 0$ implies $\lim_{t \rightarrow 0+} \frac{f(x+te_j) - f(x)}{t} = \infty$, then under some additional mild conditions, we can be sure that any solution x^* satisfies $x_j^* > 0$ for each coordinate j . (See Problem Set 1.)

Assume further that f is monotonically increasing in the sense that $f(x) > f(x')$ whenever $x_j > x'_j$ for each j , so that the consumer prefers more to less. Then, we have $w - px^* = 0$ at any solution x^* , that is, the budget constraint binds.

We can then be sure that any solution x^* to the consumer's problem with the constraint set $\mathcal{D}$ actually belongs to the subset $\mathcal{D}_0 = \{x \in \mathbb{R}^n : w - px = 0 \text{ and } x_j > 0 \text{ for } j = 1, \dots, n\}$. (See Figure 2.6.)

Note also that if $x^* \in \mathcal{D}_0$ maximizes f on the larger set $\mathcal{D}$, it maximizes f also on the smaller set $\mathcal{D}_0$. That is, x^* also solves the problem

$$\max f(x) \quad \text{s.t.} \quad x \in \mathcal{D}_0. \tag{P'}$$

Now, we have an equality constraint problem that actually has a solution! So, Step 1 of the cookbook procedure is verified.

Only one issue remains: Suppose that we found a solution x' to the problem (P'). How do we know that x' also solves the consumer's problem on the larger domain

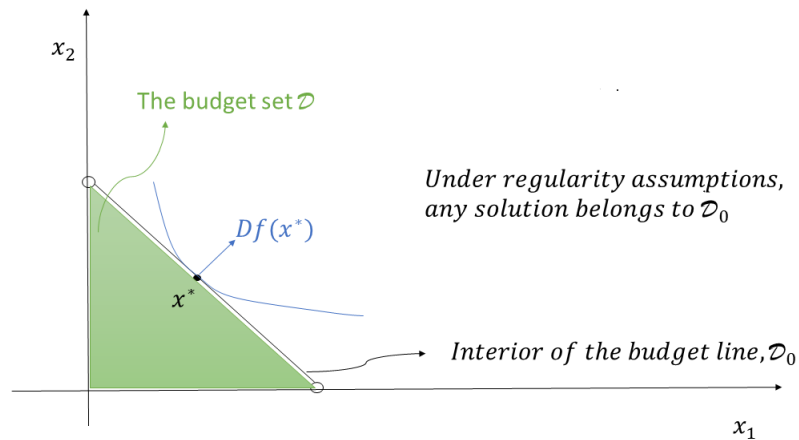


Figure 2.6: Consumer's Problem with weak Inequalities

$\mathcal{D}$? This is true because the latter problem has a solution x^* in $\mathcal{D}_0$, and that implies $f(x') \geq f(x^*)$. That is,

$$f(x') \geq f(x^*) \geq f(x) \quad \text{for every } x \in \mathcal{D}.$$

Similarly, in some other examples, it may be possible to replace a non-bounded constraint set with a bounded one to be able to use the Weierstrass theorem. (See the cost minimization problem in Problem Set 1.)